

$$G(s) = \frac{N(s)}{D(s)} = \frac{b_0(s-z_1)\cdots(s-z_m)}{a_0(s-p_1)\cdots(s-p_n)}$$

roots of $N(s)$
zeros z_i

characteristic
polynomial $D(s)$

roots of $D(s)$
poles p_i